Import Libraries

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings("ignore")
```

```
from google.colab import files
uploaded=files.upload()
```

Choose Files 2 files
**googleplaystore.csv**(text/csv) - 1360155 bytes, last modified: 3/16/2026 - 100% done
**googleplaystore_user_reviews.csv**(text/csv) - 7669276 bytes, last modified: 3/16/2026 - 100% done

```
Saving googleplaystore.csv to googleplaystore.csv
Saving googleplaystore_user_reviews.csv to googleplaystore_user_reviews.csv
```

```
apps = pd.read_csv("googleplaystore.csv")

reviews = pd.read_csv("googleplaystore_user_reviews.csv")
```

```
reviews.head()
```

| | App | Translated_Review | Sentiment | Sentiment_Polarity | Sentiment_Subjectivity |
|---|---|---|---|---|---|
| **0** | 10 Best Foods for You | I like eat delicious food. That's I'm cooking ... | Positive | 1.00 | 0.533333 |
| **1** | 10 Best Foods for You | This help eating healthy exercise regular basis | Positive | 0.25 | 0.288462 |
| **2** | 10 Best Foods for You | NaN | NaN | NaN | NaN |
| **3** | 10 Best Foods for You | Works great especially going grocery store | Positive | 0.40 | 0.875000 |
| **4** | 10 Best Foods for You | Best idea us | Positive | 1.00 | 0.300000 |

Next steps: Generate code with reviews New interactive sheet

```
apps.head()
```

| | App | Category | Rating | Reviews | Size | Installs | Type | Price | Content Rating | Genres | Last Updatec |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | Photo Editor & Candy Camera & Grid & ScrapBook | ART_AND_DESIGN | 4.1 | 159 | 19M | 10,000+ | Free | 0 | Everyone | Art & Design | January 7, 2018 |
| **1** | Coloring book moana | ART_AND_DESIGN | 3.9 | 967 | 14M | 500,000+ | Free | 0 | Everyone | Art & Design;Pretend Play | January 15 2018 |
| **2** | U Launcher Lite – FREE Live Cool Themes, Hide ... | ART_AND_DESIGN | 4.7 | 87510 | 8.7M | 5,000,000+ | Free | 0 | Everyone | Art & Design | Augus 1, 2018 |
| **3** | Sketch - Draw & Paint | ART_AND_DESIGN | 4.5 | 215644 | 25M | 50,000,000+ | Free | 0 | Teen | Art & Design | June 8 2018 |
| **4** | Pixel Draw - Number Art Coloring Book | ART_AND_DESIGN | 4.3 | 967 | 2.8M | 100,000+ | Free | 0 | Everyone | Art & Design;Creativity | June 20 2018 |

Next steps: Generate code with apps New interactive sheet

```
apps.shape
```

```
(10841, 13)
```

```
apps.columns
```

```
Index(['App', 'Category', 'Rating', 'Reviews', 'Size', 'Installs', 'Type',
       'Price', 'Content Rating', 'Genres', 'Last Updated', 'Current Ver',
       'Android Ver'],
      dtype='object')
```

```
# Dataset Info
apps.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10841 entries, 0 to 10840
Data columns (total 13 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   App             10841 non-null  object
 1   Category        10841 non-null  object
 2   Rating          9367 non-null   float64
 3   Reviews         10841 non-null  object
 4   Size            10841 non-null  object
 5   Installs        10841 non-null  object
 6   Type            10840 non-null  object
 7   Price           10841 non-null  object
 8   Content Rating  10840 non-null  object
 9   Genres          10841 non-null  object
 10  Last Updated    10841 non-null  object
 11  Current Ver     10833 non-null  object
 12  Android Ver     10838 non-null  object
dtypes: float64(1), object(12)
memory usage: 1.1+ MB
```

```
#removing wronge Columns
apps.drop(10472, inplace=True)
```

```
#Convert Reviews to Integer
apps['Reviews'] = apps['Reviews'].astype(int)
```

Clean Installs Column

```
apps['Installs'] = apps['Installs'].str.replace('+','')
apps['Installs'] = apps['Installs'].str.replace(',','')
apps['Installs'] = apps['Installs'].astype(int)
```

Clean Price Column

```
apps['Price'] = apps['Price'].str.replace('$','')
apps['Price'] = apps['Price'].astype(float)
```

```
#checking the null values
apps.isnull().sum()
```

| | 0 |
|---|---|
| App | 0 |
| Category | 0 |
| Rating | 1474 |
| Reviews | 0 |
| Size | 0 |
| Installs | 0 |
| Type | 1 |
| Price | 0 |
| Content Rating | 0 |
| Genres | 0 |
| Last Updated | 0 |
| Current Ver | 8 |
| Android Ver | 2 |

**dtype:** int64

Cleaning the Dataset

```
#Handle Rating Missing Values
apps['Rating'].fillna(apps['Rating'].mean(), inplace=True)
```

```
#Now Check Dataset Again
apps.describe()
```

| | Rating | Reviews | Installs | Price |
|---|---|---|---|---|
| count | 10840.000000 | 1.084000e+04 | 1.084000e+04 | 10840.000000 |
| mean | 4.191757 | 4.441529e+05 | 1.546434e+07 | 1.027368 |
| std | 0.478907 | 2.927761e+06 | 8.502936e+07 | 15.949703 |
| min | 1.000000 | 0.000000e+00 | 0.000000e+00 | 0.000000 |
| 25% | 4.100000 | 3.800000e+01 | 1.000000e+03 | 0.000000 |
| 50% | 4.200000 | 2.094000e+03 | 1.000000e+05 | 0.000000 |
| 75% | 4.500000 | 5.477550e+04 | 5.000000e+06 | 0.000000 |
| max | 5.000000 | 7.815831e+07 | 1.000000e+09 | 400.000000 |

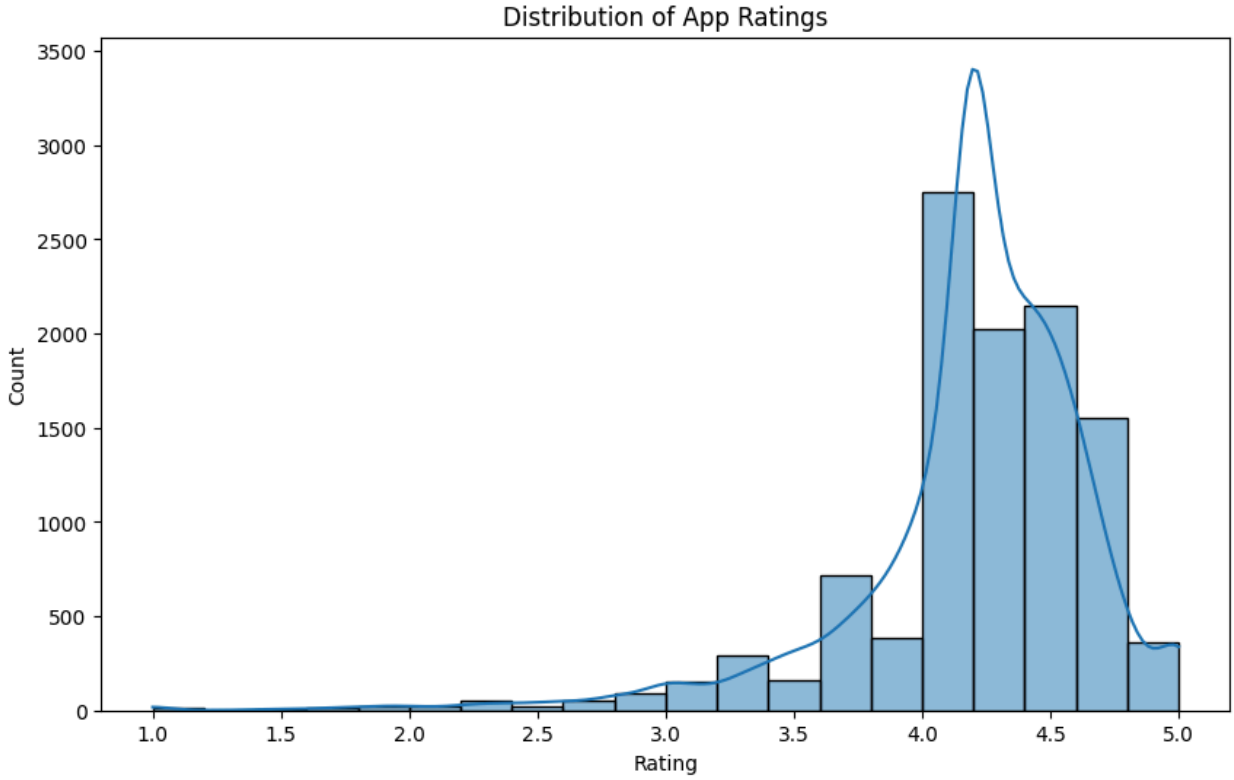
1. Rating Distribution

```
plt.figure(figsize=(10,6))
```

```
sns.histplot(apps['Rating'], bins=20, kde=True)

plt.title("Distribution of App Ratings")

plt.show()
```




Most apps have ratings between 4.0 and 4.5, indicating generally positive user feedback

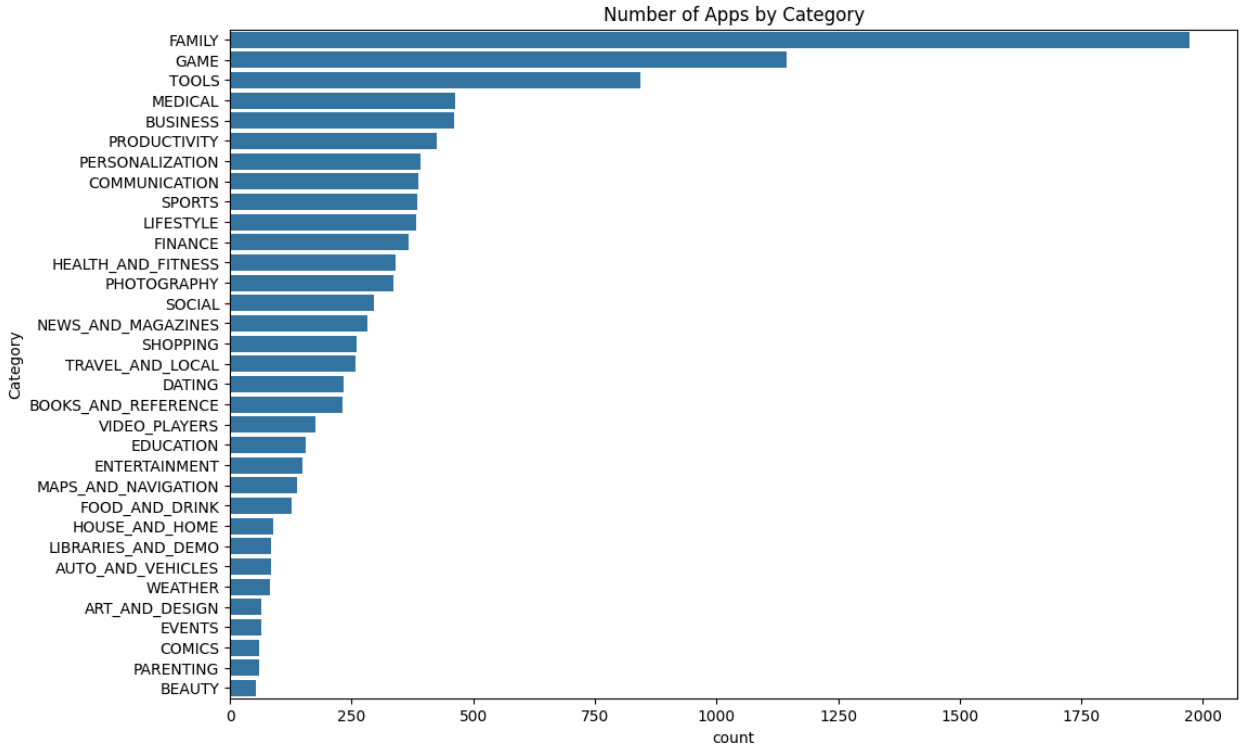
2. Apps Count by Category

```
plt.figure(figsize=(12,8))

sns.countplot(y='Category', data=apps,
order=apps['Category'].value_counts().index)

plt.title("Number of Apps by Category")

plt.show()
```




Categories such as Family, Game, and Tools contain the highest number of apps.

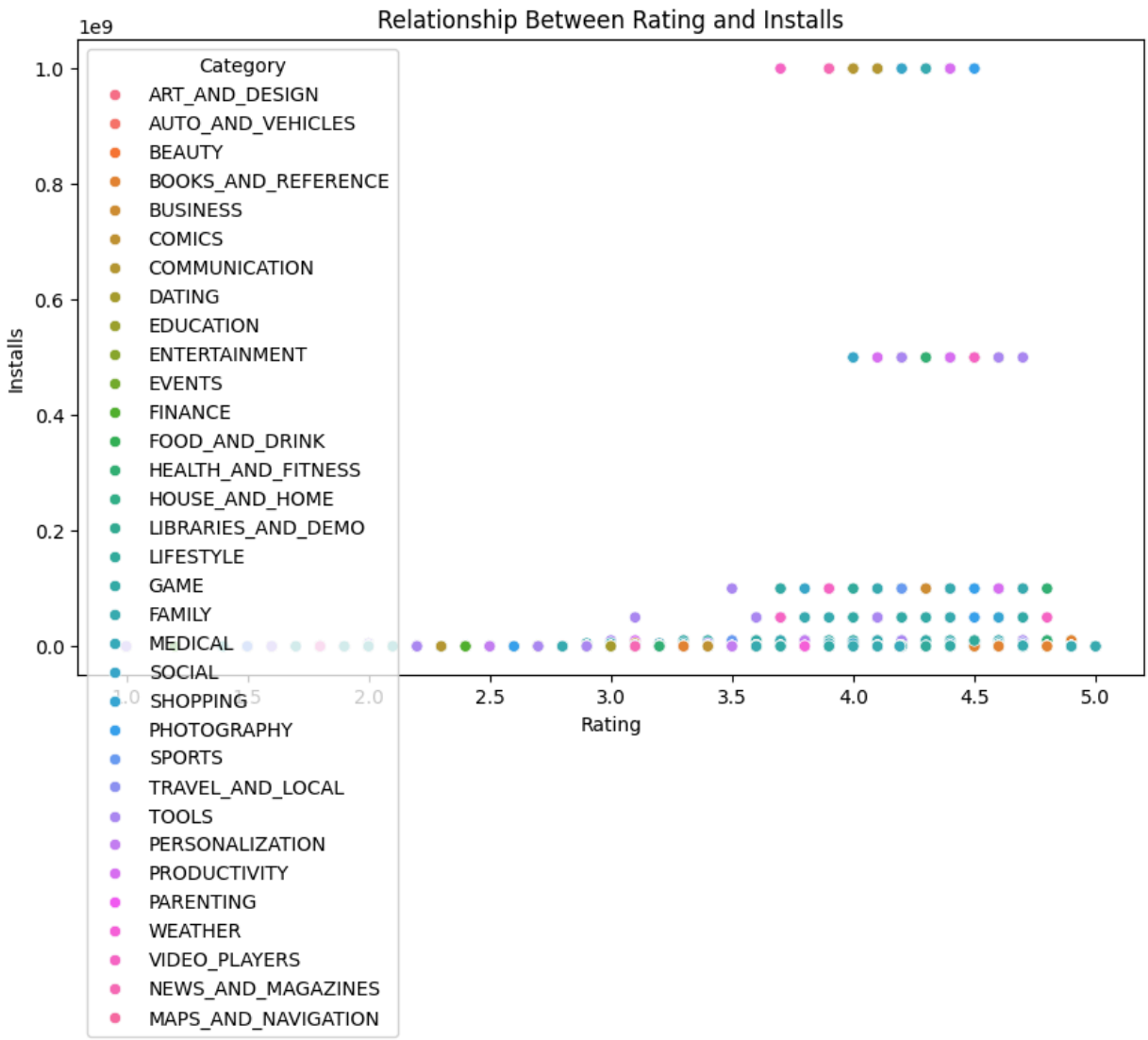
3. Relationship Between Installs and Ratings

```
plt.figure(figsize=(10,6))

sns.scatterplot(x='Rating', y='Installs',
hue='Category', data=apps)

plt.title("Relationship Between Rating and Installs")

plt.show()
```




Apps with higher ratings generally tend to have more installations, although popularity also depends on category.

4. Top 10 Most Installed Apps

```
top_apps = apps[['App','Installs']].sort_values(by='Installs', ascending=False).head(10)

top_apps
```

| | App | Installs |
|---|---|---|
| **2808** | Google Photos | 1000000000 |
| **2853** | Google Photos | 1000000000 |
| **3223** | Maps - Navigate & Explore | 1000000000 |
| **3117** | Maps - Navigate & Explore | 1000000000 |
| **3234** | Google | 1000000000 |
| **3232** | Google Street View | 1000000000 |
| **3127** | Google Street View | 1000000000 |
| **2554** | Google+ | 1000000000 |
| **464** | Hangouts | 1000000000 |
| **468** | Skype - free IM & video calls | 1000000000 |

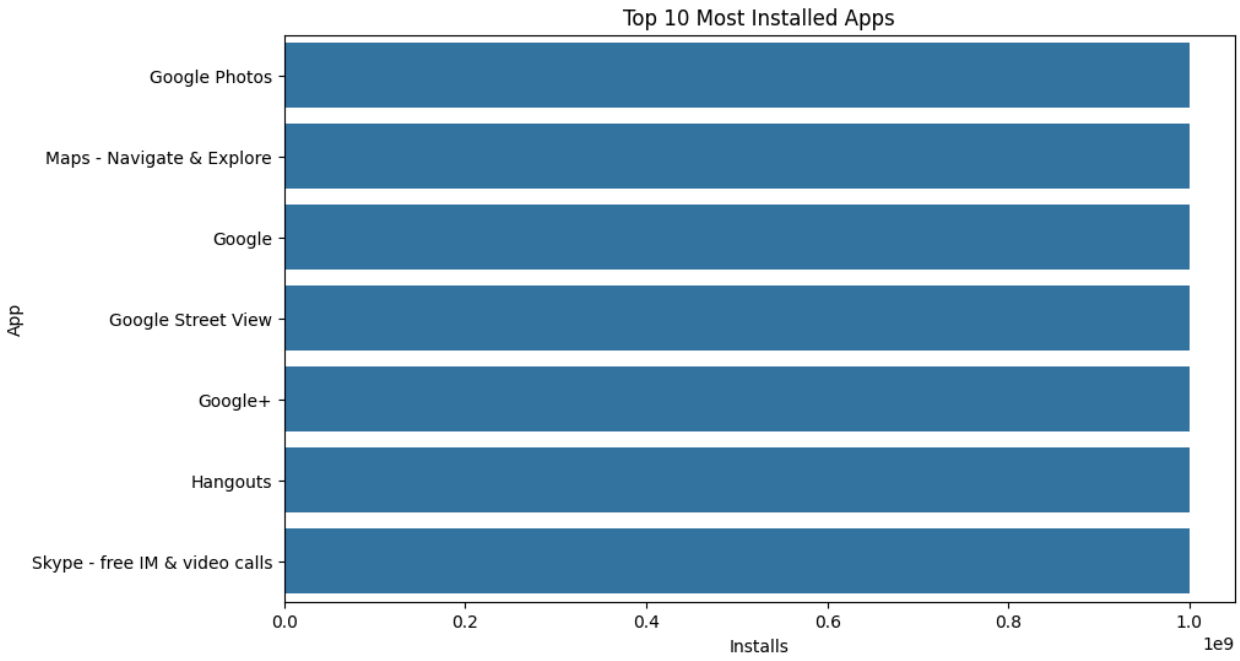
Next steps: Generate code with top_apps New interactive sheet

```
plt.figure(figsize=(10,6))

sns.barplot(x='Installs', y='App', data=top_apps)

plt.title("Top 10 Most Installed Apps")

plt.show()
```




Global apps such as Google services and social media apps dominate the install counts.

5. Top Genres

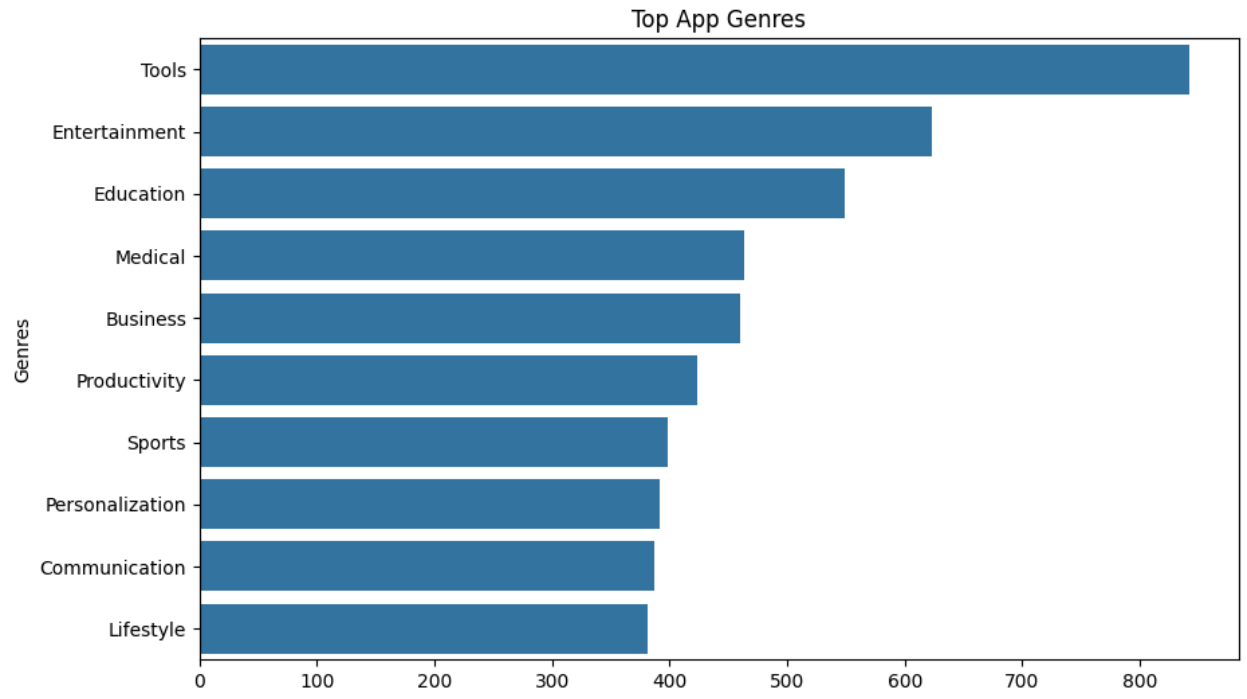
```
top_genres = apps['Genres'].value_counts().head(10)

plt.figure(figsize=(10,6))

sns.barplot(x=top_genres.values, y=top_genres.index)

plt.title("Top App Genres")

plt.show()
```




Certain genres dominate the Play Store, showing strong demand in specific categories.

6. Correlation Heatmap

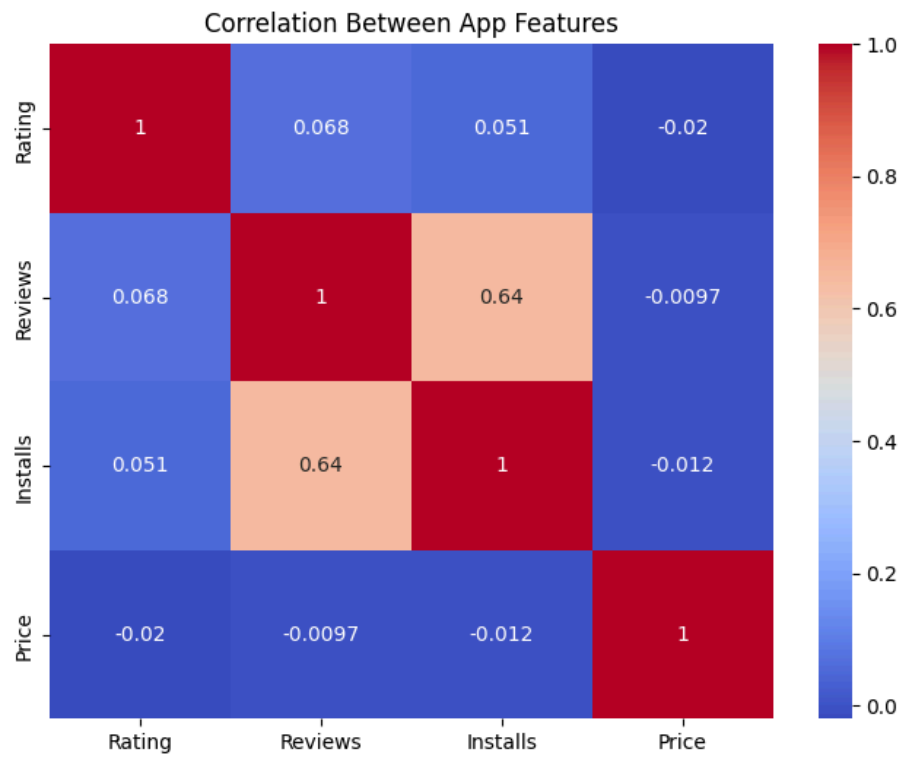
```
plt.figure(figsize=(8,6))

corr = apps.corr(numeric_only=True)

sns.heatmap(corr, annot=True, cmap="coolwarm")

plt.title("Correlation Between App Features")

plt.show()
```



The heatmap shows relationships between numerical features such as installs, reviews, and ratings.

### Check Reviews Dataset

```
reviews.head()
```

| | App | Translated_Review | Sentiment | Sentiment_Polarity | Sentiment_Subjectivity |
|---|---|---|---|---|---|
| 0 | 10 Best Foods for You | I like eat delicious food. That's I'm cooking ... | Positive | 1.00 | 0.533333 |
| 1 | 10 Best Foods for You | This help eating healthy exercise regular basis | Positive | 0.25 | 0.288462 |
| 2 | 10 Best Foods for You | NaN | NaN | NaN | NaN |
| 3 | 10 Best Foods for You | Works great especially going grocery store | Positive | 0.40 | 0.875000 |
| 4 | 10 Best Foods for You | Best idea us | Positive | 1.00 | 0.300000 |

Next steps: Generate code with reviews New interactive sheet

```
reviews.shape
```

```
(64295, 5)
```

```
reviews.isnull().sum()
```

| | 0 |
|---|---|
| App | 0 |
| Translated_Review | 26868 |
| Sentiment | 26863 |
| Sentiment_Polarity | 26863 |
| Sentiment_Subjectivity | 26863 |

**dtype:** int64

```
#Remove Missing Reviews
reviews.dropna(inplace=True)
```

```
reviews.isnull().sum()
```

| | 0 |
|---|---|
| App | 0 |
| Translated_Review | 0 |
| Sentiment | 0 |
| Sentiment_Polarity | 0 |
| Sentiment_Subjectivity | 0 |

**dtype:** int64

1. Sentiment Distribution

```
reviews['Sentiment'].value_counts()
```

| Sentiment | count |
|---|---|
| **Positive** | 23998 |
| **Negative** | 8271 |
| **Neutral** | 5158 |

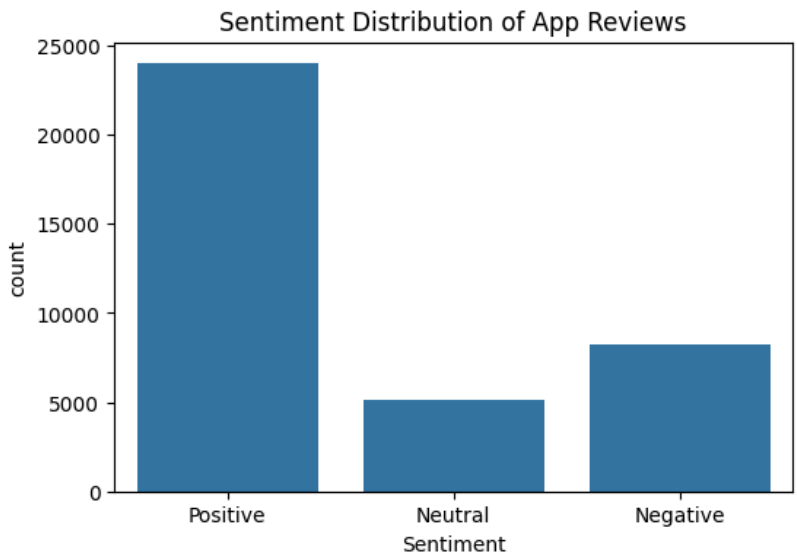
2. Sentiment Visualization

```
plt.figure(figsize=(6,4))

sns.countplot(x='Sentiment', data=reviews)

plt.title("Sentiment Distribution of App Reviews")

plt.show()
```




Most user reviews are positive, indicating that users generally have a favorable opinion about apps on the Play Store.

3. Average Sentiment by App

```
sentiment_by_app = reviews.groupby('App')['Sentiment_Polarity'].mean().sort_values(ascending=False)

sentiment_by_app.head(10)
```

| App | Sentiment_Polarity |
|---|---|
| **HomeWork** | 1.000000 |
| **Google Slides** | 0.933333 |
| **Daily Workouts - Exercise Fitness Routine Trainer** | 0.800000 |
| **Bed Time Fan - White Noise Sleep Sounds** | 0.781250 |
| **Cameringo Lite. Filters Camera** | 0.770269 |
| **Google Primer** | 0.750000 |
| **GPS Map Free** | 0.700000 |
| **GPS Speedometer and Odometer** | 0.687500 |
| **Best Ovulation Tracker Fertility Calendar App Glow** | 0.595313 |
| **3D Live Neon Weed Launcher** | 0.568182 |

**dtype:** float64

4. Merge Both Datasets